Applications of structured random matrices in probabilistic algorithms for constructing approximate matrix decompositions

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Introduction

Matrix decomposition methodologies form the cornerstone of numerical linear algebra. This research advances the theoretical framework by examining randomized algorithmic approaches. Probabilistic computation techniques offer potent mechanisms for generating approximate matrix factorizations. Our research focuses on performance optimization through the strategic implementation of structured random matrices during projection operations.

Problem Formulation

Truncated variants of QR factorization and singular value decomposition (SVD) are frequently employed to derive low-rank approximations of arbitrary matrices:

$$\mathbf{A} \approx \mathbf{B} \times \mathbf{C},$$
$$m \times n \quad m \times k \quad k \times n.$$

The computational procedure for generating low-rank matrix approximations decomposes into two distinct phases:

Phase A: Derivation of an approximate basis spanning the range space of the input matrix A. Formally, we seek a matrix Q satisfying

$$\mathbf{Q} \text{ possesses orthonormal columns and } \mathcal{A} \approx \mathbf{Q}\mathbf{Q}^{\mathbf{T}}\mathcal{A}. \qquad (1.1)$$

While column minimization in the basis matrix Q is desirable, approximation fidelity remains the paramount consideration.

Phase B: Utilizing the matrix $\mathbf{Q}$ conforming to (1.1) to facilitate computation of canonical matrix factorizations (QR, SVD, etc.) of $\mathcal{A}$.

Proto-Algorithm: Fixed-Rank Approximation Protocol

For an mn matrix $\mathcal{A}$ with target rank k and oversampling parameter p, this procedure computes an $m(k+p)$ orthonormal matrix $\mathcal{Q}$ providing an approximate basis for the range of $\mathcal{A}$.

1. Generate a random $n(k+p)$ test matrix $\mathbf{\Omega}$.
2. Compute the matrix product $\mathcal{Y} = \mathcal{A}\mathbf{\Omega}$.
3. Derive matrix $\mathcal{Q}$ through orthonormalization of $\mathcal{Y}$'s column space.

Within randomized approximation schemes, **Phase A** typically dominates the computational complexity relative to **Phase B**. The primary bottleneck in **Phase A** emerges from the matrix product $\mathcal{A}\mathbf{\Omega}$. Strategic implementation of structured random matrices $\mathbf{\Omega}$ enables optimization of this operation through algebraic reorganization.

Projection Matrix Formulation

$$\Omega = \sqrt{\frac{n}{\ell}} DFR, \qquad (1)$$

[1] where:

$$D = \begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{pmatrix}, \quad d_k \sim \mathcal{N}_{(0,1)}$$

- F represents an $n \times n$ unitary matrix with efficient matrix multiplication properties.
- R constitutes an $n \times \ell$ random subsampling operator performing uniform coordinate selection, constructed by random sampling of ℓ columns from the identity matrix.

Fast Fourier Transform Implementation

The canonical implementation employs matrices defined by:

$$f_{pq} = n^{-1/2} e^{-2\pi i (p-1)(q-1)/n}, \quad p, q = 1, 2, \ldots, n$$

Computational complexity for matrix multiplication: $O(n \log(n))$ [2]

Theoretical Proposition: Hadamard-Based Randomization

The Hadamard matrix hierarchy is recursively defined [3]:

$$H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H_q = \begin{pmatrix} H_{q/2} & H_{q/2} \\ H_{q/2} & -H_{q/2} \end{pmatrix}$$

Through computation of $H_{q/2}(\boldsymbol{x}_1 + \boldsymbol{x}_2)$ and $H_{q/2}(\boldsymbol{x}_1 - \boldsymbol{x}_2)$, we derive the recurrence relation and corresponding recursive algorithm:

$$H_q \boldsymbol{x} = \begin{pmatrix} H_{q/2} & H_{q/2} \\ H_{q/2} & -H_{q/2} \end{pmatrix} \begin{pmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \end{pmatrix} = \begin{pmatrix} H_{q/2}(\boldsymbol{x}_1 + \boldsymbol{x}_2) \\ H_{q/2}(\boldsymbol{x}_1 - \boldsymbol{x}_2) \end{pmatrix} \qquad (2)$$

Computational complexity of HDR: $O(n \log(n))$

SVD Applications

Define $B = Q^T A$.

- Compute the SVD decomposition of the reduced matrix: $B = \tilde{U}\Sigma V^T$.
- Construct $U = Q\tilde{U}$

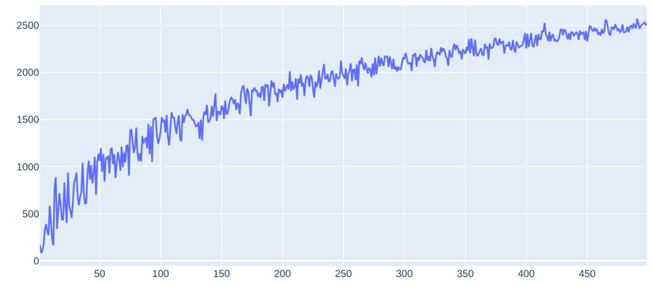
Given a matrix satisfying condition (1.1), we demonstrate that the matrices U, Σ, and V^T constitute a valid SVD decomposition for matrix A.

The primary limitation occurs when the column dimension of $\mathcal{Q}$ is insufficient to span the rank of matrix $\mathcal{A}$. Adequate column dimension ensures comprehensive directional coverage through projections.
Regarding SVD approximation, the following error bounds [1] hold:

$$\mathbb{E}\|A - QQ^*A\| \leq \left[1 + \frac{4\sqrt{k+p}}{p-1} \cdot \sqrt{\min\{m,n\}}\right] \sigma_{k+1} \qquad (3)$$

Error Correlation Analysis

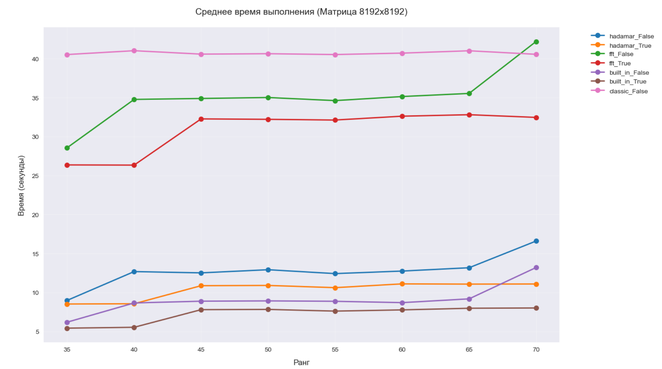
The functional relationship between approximation error and σ_{k+1} is visually demonstrated:



Rank Estimation Algorithm

Constructing Q according to the prescribed algorithm, the condition $||A - QQ^*A|| < \epsilon$ indicates satisfactory approximation, implying rank(A) $\leq$ rank(Q) $= (k+p)$. This property facilitates binary search implementation for rank estimation through iterative error evaluation.

Comparative computational complexity across projection methodologies:



References

[1] Joel A Tropp Nathan Halko, Per-Gunnar Martinsson. *Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions*. Society for Industrial and Applied Mathematics, 2011.

[2] Tukey J. W. Cooley J. W. An algorithm for the machine calculation of complex fourier series. (19), 1965.

[3] Steven Zucker Edo Liberty. Accelerated dense random projections. 2009.